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What Is the Impact of Introducing a Parallel OTC Market? Theory and Evidence from the Chinese Interbank FX Market

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1 Big picture

- **Question.** What happens when a decentralized over-the-counter (OTC) market is introduced in parallel to an existing centralized anonymous limit order book (LOB)? Which venue dominates, how do transaction-cost schedules change with trade size, and do all traders face the same price function?
- **Setting.** The paper studies the Chinese interbank foreign exchange market, where spot USD/CNY trading was originally conducted entirely on a centralized LOB run by the China Foreign Exchange Trading System (CFETS). On January 4, 2006, regulators introduced a parallel OTC market while keeping the LOB open.
- **Core challenge.** In most markets we do not observe two very different trading mechanisms coexisting for the *same asset* in the *same market* after a clean institutional reform. This event lets the authors separate “asset-class effects” from “market-design effects.”
- **Main empirical result.** Trading rapidly migrated to the OTC market. By July 2006, **89.5%** of trades and **99.0%** of dollar volume were executed OTC.
- **Price-function result.** The LOB has an **upward-sloping** price function: bigger trades face larger transaction costs. The OTC has a **downward-sloping** price function: bigger trades receive lower transaction costs.
- **Clientele result.** The LOB effectively has a **single price function**, while the OTC has **multiple price functions**: large banks get the best prices, medium banks intermediate prices, and small banks the worst prices.
- **Methodological contribution.** The paper combines reduced-form evidence, a Heckman selection model, and a structural theory of **parallel LOB and OTC markets**. The theory nests adverse selection in the anonymous LOB and bargaining/search in the non-anonymous OTC.
- **New prediction from the model.** Traders with more bargaining power should switch from the LOB to the OTC at *smaller* trade sizes.
- **Test of the new prediction.** During the first six months after the reform, the switching points are roughly **\$3 million** for large banks, **\$10 million** for medium banks, and **\$20 million** for small banks. After the transition, those thresholds fall to about **\$0.5 million**, **\$5 million**, and **\$10 million**.
- **Economic takeaway.** Market design matters a lot. The same asset can have an upward-sloping price schedule in an anonymous adverse-selection market and a downward-sloping, clientele-specific price schedule in a decentralized bargaining market.

2 Data and measurement (key objects)

2.1 Institutional setting and the policy change

- **Initial market structure.** Before 2006, the Chinese interbank FX market was organized as a **nationwide centralized LOB** run by the CFETS. Orders were anonymous and matched under **price priority first** and then **time priority**.
- **Parallel OTC introduction.** On **January 4, 2006**, the People’s Bank of China introduced a **parallel OTC market**. The reform was designed to:
 - accommodate rapidly growing FX volume,
 - move the market structure closer to international FX practice,
 - enhance liquidity and price discovery,
 - do so in a gradual and controllable way by *keeping the LOB open*.
- **Why this institutional change is useful.** Because the same currency pair can be traded in either venue after the reform, the paper can compare outcomes across two mechanisms without changing the underlying asset.
- **Who trades in OTC?** In the OTC, market participants are customers and dealers. A dealer posts two-sided quotes and bilateral trades are executed through the CFETS infrastructure. Customer-dealer trades account for about **60%** of total trade volume, and interdealer trades for about **40%**.
- **Why credit-risk differences are not the main story.** On the LOB, CFETS guarantees trades. In OTC, counterparties bear credit risk directly. But the paper argues that spot FX credit risk is minimal because:
 - settlement is short horizon (mostly **T+2**),
 - spot FX is unlevered relative to derivatives,
 - dealers report that only a tiny portion of bilateral credit lines is devoted to spot FX,
 - dealers report very low intraday default concerns.

2.2 Transaction data and participant classification

- **Source.** Intraday trade records from the **China Foreign Exchange Trading System (CFETS)**.
- **Sample period.** **August 2005 to December 2006**, giving:
 - **five months before** OTC introduction,
 - **twelve months after** OTC introduction.
- **Currency pair.** The paper focuses on **USD/CNY spot trades**, which account for more than **90%** of spot volume in the Chinese interbank FX market.
- **What each record contains.**

- time of transaction,
 - trade size,
 - exchange rate,
 - whether the trade is buyer-initiated, seller-initiated, or interdealer,
 - a unique participant identifier.
- **Bank-size classification.** CFETS classifies participants as:
 - **core banks** (interpreted as large banks),
 - **fundamental banks** (medium banks),
 - **ordinary banks** (small banks).

The paper uses this categorization as a proxy for market importance and bargaining power.

- **Trade counts.** Table 1 reports **30,778** LOB customer-dealer trades in the August–December 2005 pre-period and, in 2006, **15,097** LOB trades versus **87,407** OTC trades.

2.3 Main empirical objects

- **Trade-size buckets.** Trades are divided into seven buckets:

$$< \$0.5\text{m}, 0.5\text{--}1\text{m}, 1\text{--}3\text{m}, 3\text{--}5\text{m}, 5\text{--}10\text{m}, 10\text{--}20\text{m}, > \$20\text{m}.$$

- **Disaggregated versus aggregated trades.** The authors worry that large parent orders may be split into smaller child orders, especially on the LOB. So they construct an **aggregated trade** by combining all trades from the same bank, on the same side, within **ten minutes**. Their main results are robust, and later analysis uses aggregated trades.
- **Benchmark for transaction costs.** Because quoted bid and ask data are unavailable, the paper constructs an **interdealer bid-ask midpoint** from the most recent interdealer liquidity-demanding buy and sell.
- **Main transaction-cost measure.**

$$\text{Relative Effective Spread}_k = D_k(\ln(P_k) - \ln(M_k)),$$

where:

- $D_k = +1$ for a buyer-initiated customer-dealer trade,
 - $D_k = -1$ for a seller-initiated customer-dealer trade,
 - P_k is the trade price,
 - $M_k = (B_k + A_k)/2$ is the interdealer midpoint just before the trade,
 - B_k and A_k are the most recent interdealer traded bid and ask.
- **Why this measure is nice.** It gives a trade-level proxy for one-way transaction costs even without quoted order-book data. Commissions and fees are embedded in prices in both venues, so the measure captures total trading cost.

- **Venue-choice variable.** In the multivariate analysis, a binary variable indicates whether a trade is executed in the OTC or the LOB.

2.4 Key descriptive facts and selection patterns

- **OTC migration is fast.** Over 2006 as a whole, the OTC handles **85.3%** of trade count and **95.1%** of dollar volume.
- **Large trades migrate more.** After the transition period, the OTC share of *aggregated* trade counts is:
 - **52.8%** for trades below \$0.5 million,
 - **71.2%** for \$0.5–\$1 million,
 - **77.6%** for \$1–\$3 million,
 - **95.2%** for \$3–\$5 million,
 - **99.2%** for \$5–\$10 million,
 - **96.7%** for \$10–\$20 million,
 - **98.1%** for above \$20 million.
- **Bank composition of OTC changes.** Between January and December 2006, the share of large banks in OTC trading rises from about **40% to 60%**, the share of medium banks falls from **60% to 38%**, and the share of small banks stays around **2%**.
- **Selection issue.** This already suggests strong endogeneity in venue choice:
 - large trades choose OTC,
 - large banks choose OTC,
 - small trades and small banks are relatively more likely to stay in the LOB.

2.5 Unobservables and timing

- **Unobserved heterogeneity.**
 - bargaining power is not directly observed and is proxied by bank category,
 - bilateral OTC credit lines are not directly observed,
 - informed trading is not directly observed but is central to the theory.
- **Empirical timing.** The reform creates a clean pre-period (Aug–Dec 2005), a transition period (Jan–Jun 2006), and a post-transition period (Jul–Dec 2006).
- **Theoretical timing.**
 1. At time t , liquidity suppliers post limit orders in the LOB.
 2. At time $t + 1$, either an informed or an uninformed liquidity demander arrives and chooses order size and possibly venue.
 3. At time $t + 2$, the fundamental value of the currency is realized.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing empirical hypotheses

The paper organizes the analysis around six hypotheses.

- **Hypothesis 1.** The LOB should predominate, because LOBs are thought to have lower transaction costs and more efficient price-quantity schedules.
- **Hypothesis 2.** The OTC should predominate, because large institutional traders may be able to bargain for better prices there.
- **Hypothesis 3.** The slope of the price function is determined by the **asset class**. Since both venues trade FX, both price functions should slope the same way.
- **Hypothesis 4.** The slope of the price function is determined by the **trading mechanism**. Then the LOB should be upward-sloping and the OTC downward-sloping.
- **Hypothesis 5.** The LOB should have a single price function, while the OTC should have multiple price functions because counterparties can bargain differently with different clienteles.
- **Hypothesis 6.** Competition among OTC dealers is so strong that both venues should have a single price function.
- **Interpretation.** The empirical design is neat because the same event speaks simultaneously to venue dominance, the slope of price functions, and clientele-specific pricing.

3.2 Effective-spread measure and the selection model

The venue-choice equation is a Heckman first-stage Probit:

$$Y = \Phi(\text{Bank}, \text{TradeSize}, \text{Transition}, \text{Controls}, \varepsilon),$$

where $Y = 1$ if the trade is executed OTC and $Y = 0$ if it is executed in the LOB.

The second-stage spread regressions are venue-specific:

$$\text{RES}_i^v = \alpha_v + \beta_{1,v} \text{Medium}_i + \beta_{2,v} \text{Large}_i + \gamma_{1,v} \text{Size}_i + \gamma_{2,v} \text{Size}_i^2 + \delta_v X_i + \lambda_v \widehat{\text{IMR}}_i + u_i,$$

for venue $v \in \{\text{OTC}, \text{LOB}\}$.

Key regressors:

- bank-size dummies (medium and large, with small as omitted),
- trade size and trade size squared,
- market conditions: hourly exchange-rate return and realized volatility,
- transition dummy for the first six months after the reform,
- buy dummy,

- inverse Mills ratio from the first-stage Probit.
- **Interpretation.** The first stage models the endogenous sorting of trades into OTC versus LOB. The second stage then asks:
 - is the slope positive or negative within each venue?
 - do large and medium banks face systematically different price schedules?

3.3 Centralized LOB model

There are three classes of agents:

- many risk-neutral uninformed liquidity suppliers,
- one risk-neutral informed liquidity demander,
- uninformed liquidity demanders with exogenous liquidity needs.

The time- $t + 2$ foreign-currency value is discrete:

$$v \in \{v_1, v_2, \dots, v_K\}, \quad \Pr(v = v_k) = \pi_k.$$

At time $t + 1$, with probability ρ the arriving trader is informed; otherwise the arriving trader is uninformed.

If informed, the trader's signal is $s = v$, and the optimal market buy is

$$x = \sum_{n=1}^{N+1} Q_n I_n,$$

where $I_n = 1$ if $p_n < v$ and $I_n = 0$ otherwise.

The informed trader's profit is

$$\sum_{n=1}^{N+1} (v - p_n) Q_n I_n.$$

The optimal size of the limit sell posted at ask price p_n is

$$Q_n^* = \frac{\rho 2M \sum_{k=1}^K \pi_k (v_k - p_n) I_k}{(1 - \rho)(E[v] - p_n)} + M - \sum_{i=1}^{n-1} Q_i, \quad n = 1, 2, \dots, N,$$

and

$$Q_{N+1}^* = \frac{M - \sum_{i=1}^N p_i Q_i}{p_{N+1}}.$$

- **Interpretation.** The centralized LOB is an **anonymous adverse-selection market**. Larger orders are more likely to come from informed traders with stronger signals. Liquidity suppliers therefore charge more for deeper execution. This is what makes the LOB **upward-sloping**.

3.4 Decentralized OTC model

In the OTC, counterparties know who they trade with. The paper argues that informed traders would eventually be blacklisted in such a repeated non-anonymous environment, so equilibrium OTC trading is built around **uninformed** liquidity demanders and liquidity suppliers.

There are J classes of liquidity demanders indexed by $j = 1, 2, \dots, J$. Let ϕ_j denote the *liquidity supplier's* bargaining power when facing a type- j trader. Lower ϕ_j means more bargaining power for the customer.

Liquidity demanders have reservation prices

$$E[v] + C \quad \text{for buys,} \quad E[v] - C \quad \text{for sells,}$$

and liquidity suppliers incur a fixed order-processing cost f per trade.

The equilibrium OTC price for a trade of size U is

$$p(U) = E[v] + I \frac{f}{|U|} + I \phi_j \left(C - \frac{f}{|U|} \right),$$

where $I = +1$ for a buy and $I = -1$ for a sell.

- **Interpretation.**

- The term $f/|U|$ creates a **fixed-cost dilution effect**: larger trades spread the fixed processing cost over more units.
- That yields a **downward-sloping OTC price function**.
- Heterogeneity in ϕ_j yields **multiple OTC price functions**: powerful traders get better prices.

3.5 Parallel markets and the choice of venue

To combine the two venues, the paper assumes:

- informed traders trade only in the LOB,
- liquidity suppliers operate in both venues,
- uninformed liquidity demanders are split into:
 - **nondiscretionary** traders who remain in the LOB,
 - **discretionary** traders who choose between LOB and OTC.

A discretionary trader of type j compares the LOB execution price with the OTC negotiated price and chooses the cheaper venue.

The critical switching quantity for a trader of type j whose LOB order would finish at price p_n is

$$S_{j,n}^* = \frac{-b_{j,n} + \sqrt{b_{j,n}^2 - 4a_{j,n}c_{j,n}}}{2a_{j,n}},$$

where

$$\begin{aligned} a_{j,n} &= p_n^2 - E[v]p_n - I\phi_j C p_n, \\ b_{j,n} &= 2d_n p_n - E[v](d_n + p_n e_n) - I\phi_j (C d_n + C p_n e_n - f) - I f, \\ c_{j,n} &= d_n^2 - E[v]d_n e_n - I\phi_j e_n (C d_n - f) - I f d_n, \end{aligned}$$

and

$$d_n = \sum_{i=1}^{n-1} p_i Q_i, \quad e_n = \sum_{i=1}^{n-1} Q_i.$$

The optimal venue is:

$$\text{LOB if } |U| < \sum_{i=1}^{n-1} Q_i + S_{j,n}^*, \quad \text{OTC if } |U| > \sum_{i=1}^{n-1} Q_i + S_{j,n}^*.$$

- **Interpretation.** The switching quantity is the point at which the trader is indifferent between:
 - anonymous, adverse-selection pricing in the LOB,
 - non-anonymous, bargaining-based pricing in the OTC.

Small orders may stay in the LOB, while large orders switch to OTC.

3.6 Parallel-market equilibrium and the new comparative static

When some discretionary traders switch to OTC, there are fewer uninformed traders left in the LOB to camouflage informed traders. That increases the conditional probability of informed trading in the LOB beyond the switching point.

The revised optimal limit-sell size becomes

$$\hat{Q}_n^* = \frac{\rho_j 2M \sum_{k=1}^K \pi_k (v_k - p_n) I_k}{(1 - \rho_j)(E[v] - p_n)} + M - \sum_{i=1}^{n-1} Q_i, \quad n = 1, 2, \dots, N,$$

with

$$\hat{Q}_{N+1}^* = \frac{M - \sum_{i=1}^N p_i Q_i}{p_{N+1}}.$$

The paper's key comparative-static result is:

$$S_{j,n}^* \text{ is strictly increasing in } \phi_j.$$

- **Interpretation.** Remember that ϕ_j is the liquidity supplier's bargaining power. So when the *customer* has **more** bargaining power, ϕ_j is smaller and the switching threshold is lower. Hence:

large banks switch to OTC at smaller trade sizes.

3.7 Numerical illustration

The paper calibrates the model with:

$$v_1 = 6.77, \quad v_K = 6.79, \quad M = \$50 \text{ million}, \quad K = 80,$$

$$f = 0.320, \quad C = 0.009, \quad \rho = 0.10, \quad \delta = 0.001,$$

$$\phi_1 = 0.03, \quad \phi_2 = 0.24, \quad \phi_3 = 0.32.$$

- **Interpretation.** The goal is not precise structural estimation here. Instead, the calibration shows that a single theoretical environment can qualitatively match:
 - one upward-sloping LOB price function,
 - three downward-sloping OTC price functions,
 - lower OTC switching thresholds for larger banks.

4 Identification and empirical credibility (what makes the evidence convincing)

4.1 Why this is a strong natural experiment

- The same asset (USD/CNY spot FX) trades in two venues after the reform.
- The reform is institutional rather than asset-specific: the OTC is added, but the underlying currency pair does not change.
- Traders can freely move across venues, so venue migration is economically meaningful.
- **Interpretation.** Because the asset is fixed and only the mechanism changes, the paper can ask whether the price schedule is driven by *asset class* or by *market structure*. This is the paper's cleanest identification insight.

4.2 How venue dominance is identified

- Venue dominance is identified directly from pre/post changes in trade counts and dollar volume.
- The fact that OTC dollar-volume share exceeds OTC trade-count share shows that large orders migrate faster than small ones.
- Size-specific migration is observed both in raw trades and in ten-minute aggregated trades.
- **Interpretation.** The dominance result is not just that OTC gets popular; it is that *the most economically important trades* move first.

4.3 How the slope and multiplicity of price functions are identified

- The interdealer traded bid-ask midpoint provides a benchmark for true value and allows transaction costs to be measured consistently across venues.

- Because both venues trade the same currency pair at the same time, differences in how cost varies with trade size identify the slope of each venue’s price function.
- Bank-size heterogeneity identifies whether there are single or multiple price functions within a venue.
- **Interpretation.**
 - If all bank classes line up in one venue, that venue has a single price function.
 - If large, medium, and small banks occupy different curves, that venue has multiple price functions.

4.4 Why the Heckman correction matters

- Large trades disproportionately go to OTC.
- Large banks disproportionately go to OTC.
- So raw spread comparisons mix **pricing** with **selection**.
- **Interpretation.** The Heckman model is crucial because otherwise one might wrongly attribute the low OTC costs for large trades entirely to venue design, when some of it could reflect who chooses OTC. The selection correction asks whether the venue patterns survive after controlling for endogenous sorting. They do.

4.5 Why bank size is a plausible bargaining-power proxy

- CFETS classifies core, fundamental, and ordinary banks by market importance and influence.
- Larger banks trade more, are more central to the market, and plausibly have stronger bargaining positions vis-à-vis dealers.
- The empirical ordering of OTC prices across bank groups is exactly what the bargaining interpretation predicts.

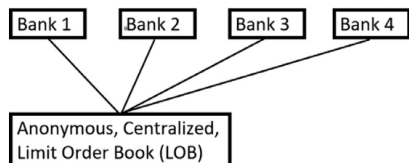
4.6 Limitations and caveats

- The paper does not observe institution names or bilateral OTC credit lines directly.
- The theory of OTC exclusion of informed traders is motivated by repeated reputational logic, but that reputational process is not explicitly modeled.
- Bank size is an indirect proxy for bargaining power rather than a direct measure.
- Some venue-choice frictions may remain beyond what the Heckman approach captures.
- **Why the paper is still persuasive.** The authors address the main OTC-credit-line concern by surveying dealers, showing near-universal coverage, and by dropping the 16 participants who never use OTC in a robustness check, with similar results.

5 Tables and figures

5.1 Figure 1 and Table 1: the reform and venue migration

Before the introduction of a parallel OTC market



After the introduction of a parallel OTC market

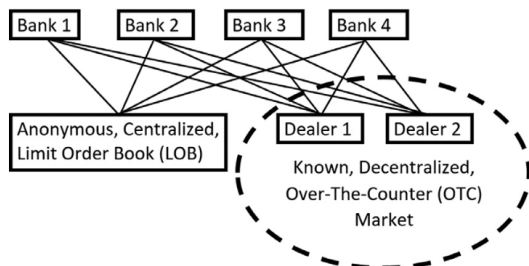


Fig. 1. Trading mechanisms in the Chinese Interbank FX market. *Note:* The labels “Bank 1”, “Bank 2”, etc. are meant to be inclusive of all market participants. The large majority of market participants are banks, but a small minority are nonbanks.

Table 1
Number of trades and OTC shares over time.
This table shows the number of trades in the Limit Order Book (LOB) and Over-The-Counter (OTC) markets, the OTC share of trades, and OTC share of dollar volume for Chinese Yuan trades against the US dollar in the Chinese Interbank Foreign Exchange market. All computations are based on China Foreign Exchange Trading System (CFETS) intraday transaction data from August 2005 to December 2006. The t-statistic for a given month is the total for that month.

	LOB	OTC	OTC%		T-test of Venue Dif	
	Trades	Trades	Trade %	Volume %	Trades	\$ Volume
Aug 2005	6,060		0.0%	0.0%		
Sep 2005	6,024		0.0%	0.0%		
Oct 2005	5,407		0.0%	0.0%		
Nov 2005	6,086		0.0%	0.0%		
Dec 2005	6,301		0.0%	0.0%		
Jan 2006	2,342	1,451	36.3%	74.0%	11.27	-5.90
Feb 2006	1,624	2,088	56.2%	88.3%	-2.71	-10.03
Mar 2006	1,610	3,996	71.3%	96.8%	-16.32	-19.54
Apr 2006	1,416	4,378	75.6%	94.3%	-13.59	-14.20
May 2006	1,054	3,971	79.0%	96.9%	-20.46	-22.68
Jun 2006	918	5,710	86.1%	98.5%	-16.79	-14.51
Jul 2006	907	7,709	89.3%	99.0%	-29.64	-31.01
Aug 2006	1,045	10,325	90.8%	99.1%	-28.16	-23.27
Sep 2006	1,002	11,725	92.1%	99.2%	-30.57	-23.94
Oct 2006	1,103	10,015	90.1%	97.9%	-41.16	-36.89
Nov 2006	1,006	12,395	92.5%	93.5%	-41.73	-14.05
Dec 2006	870	13,644	94.0%	89.6%	-29.45	-9.69
All of 2005	30,778		0.0%	0.0%		
All of 2006	15,097	87,407	85.3%	95.1%		
Full sample	45,875	87,407				

Read:

- Figure 1 is the institutional diagram. It shows that after January 2006 every bank can choose either the centralized LOB or the decentralized OTC market and can move back and forth across venues.
- Table 1 gives the main market-share result. By **July 2006**, the OTC executes **89.5%** of trade counts and **99.0%** of dollar volume.
- Over all of 2006, OTC accounts for **85.3%** of trade count and **95.1%** of dollar volume.
- The fact that OTC volume share exceeds OTC trade share means large trades migrate especially strongly to OTC.
- This table is the cleanest evidence against the “LOB is inevitable” view and in favor of the idea that large institutions prefer OTC bargaining.

5.2 Table 2, Figure 2, Figure 3, and Figure 4: migration by trade size and bank class

Read:

- Table 2 shows the migration is highly size-dependent. After the transition, OTC handles only

Table 2

Number of trades and volume share by trade size.

This table shows the number of trades and trade counts share (dollar volume share) of trades by trade size in the Limit Order Book (LOB) and Over-The-Counter (OTC) markets for Chinese Yuan trades against the US dollar in the Chinese Interbank Foreign Exchange market. All computations are based on China Foreign Exchange Trading System (CFETS) intraday transaction data from August 2005 to December 2006.

	LOB	OTC							
	Trades	Trades	Trade %	Volume %			T-test of Venue Dif		
					Trades	\$ Volume			
Panel A. Disaggregated trades before parallel OTC introduction (Aug 2005–Dec 2005)									
< 0.5 mil	9,772		0.0%	0.0%					
0.5–1 mil	4,694		0.0%	0.0%					
1–3 mil	5,551		0.0%	0.0%					
3–5 mil	3,160		0.0%	0.0%					
5–10 mil	3,775		0.0%	0.0%					
10–20 mil	2,392		0.0%	0.0%					
> 20 mil	1,434		0.0%	0.0%					
Panel B. Disaggregated trades after parallel OTC transition (Jul 2006–Dec 2006)									
< 0.5 mil	8,669	7,676	47.0%	53.7%	2.85	–3.82			
0.5–1 mil	2,985	6,501	68.5%	69.5%	–15.57	–15.68			
1–3 mil	2,350	8,090	77.5%	79.1%	–25.95	–26.26			
3–5 mil	637	13,100	95.4%	95.8%	–13.16	–13.00			
5–10 mil	345	47,094	99.3%	99.4%	–15.31	–15.24			
10–20 mil	71	2,643	97.8%	97.8%	–4.97	–4.90			
> 20 mil	40	2,303	98.3%	82.8%	–3.63	–2.77			
Panel C. Aggregated trades before parallel OTC introduction (Aug 2005–Dec 2005)									
< 0.5 mil	8,795		0.0%	0.0%					
0.5–1 mil	4,272		0.0%	0.0%					
1–3 mil	5,330		0.0%	0.0%					
3–5 mil	3,062		0.0%	0.0%					
5–10 mil	3,678		0.0%	0.0%					
10–20 mil	2,411		0.0%	0.0%					
> 20 mil	1,494		0.0%	0.0%					
Panel D. Aggregated trades after parallel OTC transition (Jul 2006–Dec 2006)									
< 0.5 mil	6,867	7,676	52.8%	59.6%	–2.73	–9.63			
0.5–1 mil	2,635	6,501	71.2%	72.2%	–18.06	–19.31			
1–3 mil	2,335	8,090	77.0%	79.1%	–26.33	–28.88			
3–5 mil	666	13,100	95.2%	95.7%	–13.81	–13.80			
5–10 mil	387	47,094	99.2%	99.4%	–16.61	–16.48			
10–20 mil	91	2,643	98.7%	97.3%	–5.56	–5.74			
> 20 mil	44	2,303	98.1%	82.5%	–3.41	–2.77			

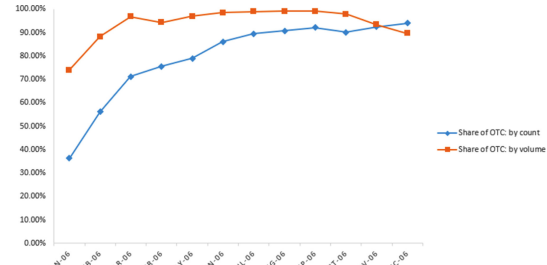


Fig. 2. The share of the OTC trades by extensive and intensive margins, 2006.

about **52.8%** of aggregated trades below \$0.5 million, but around **95%–99%** of aggregated trades above \$3 million.

- Figure 2 shows that OTC share rises rapidly in both extensive margin (trade count) and intensive margin (dollar volume) throughout 2006.
- Figure 3 makes the size heterogeneity vivid: trades above **\$3 million** are already overwhelmingly OTC early in the year, while the smallest trades migrate much more slowly.
- Figure 4 shows clientele composition inside OTC. Large-bank share rises from about **40%** to **60%**; medium-bank share falls from about **60%** to **38%**; small banks remain a tiny fraction.
- Together these figures say that the OTC is not just gaining market share mechanically. It is especially attracting the largest trades and the strongest clients.

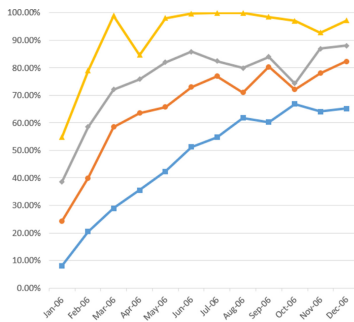


Fig. 3. The share of different trade sizes in OTC, 2006.

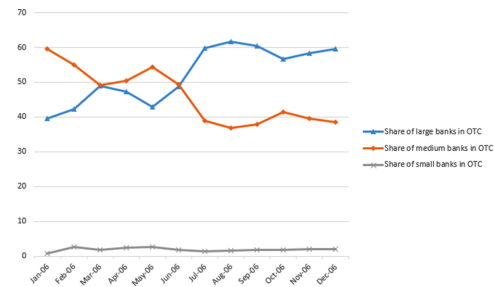


Fig. 4. Bank group market share in the OTC, 2006.

5.3 Table 3 and Table 4: the slope of the price function is about the mechanism

Read:

- Table 3 reports monthly one-way relative effective spreads. In early 2006 the LOB is cheaper than OTC, but by late 2006 the OTC becomes cheaper. For example, in **November 2006** the LOB spread is **4.92 bps** versus **2.80 bps** in OTC, and in **December 2006** it is **5.16** versus **3.53**.

- Table 4 is one of the most important tables in the paper. During the transition:
 - LOB spreads rise from **1.79** bps for trades below \$0.5 million to **9.20** bps above \$20 million.
 - OTC spreads fall from **6.51** bps to **1.65** bps across the same size range.
- After the transition, the same qualitative pattern remains:
 - LOB: **3.57** to **9.57** bps,
 - OTC: **6.23** to **0.68** bps.
- This is the core evidence against Hypothesis 3. Since both venues trade the *same FX asset*, the opposite slopes are very hard to explain by asset class. They instead point strongly to market-design effects.

Table 3

One-way relative effective spread over time.

This table shows the one-way relative effective spread in basis points (bps) over time in the Limit Order Book (LOB) and Over-The-Counter (OTC) market for Chinese Yuan trades against the US dollar in the Chinese Interbank Foreign Exchange market. All computations are based on China Foreign Exchange Trading System (CFETS) intraday transaction data from August 2005 to December 2006. The *t*-statistic for a given month is the total for that month.

	LOB (bps)	OTC (bps)	T-test
Aug 2005	2.24		
Sep 2005	1.87		
Oct 2005	1.29		
Nov 2005	0.97		
Dec 2005	0.80		
Jan 2006	2.94	3.55	-1.25
Feb 2006	1.15	4.09	-6.24
Mar 2006	2.11	3.60	-3.78
Apr 2006	3.67	5.10	-2.04
May 2006	4.22	6.25	-2.66
Jun 2006	3.50	4.54	-1.86
Jul 2006	3.73	3.59	0.22
Aug 2006	3.79	4.17	-0.63
Sep 2006	4.19	4.82	-0.71
Oct 2006	3.57	3.52	0.11
Nov 2006	4.92	2.80	4.71
Dec 2006	5.16	3.53	3.39

Table 4

One-way relative effective spread by trade size.

This table shows the one-way relative effective spread in basis points (bps) by trade size in the Limit Order Book (LOB) and Over-The-Counter (OTC) markets for Chinese Yuan trades against the US dollar in the Chinese Interbank Foreign Exchange market. All computations are based on China Foreign Exchange Trading System (CFETS) intraday transaction data from January 2006 to December 2006.

	LOB (bps)	OTC (bps)	T-test
Panel A. During the parallel OTC transition (Jan 2006–Jun 2006)			
< 0.5 mil	1.79	6.51	-8.31
0.5–1 mil	2.13	5.87	-4.67
1–3 mil	2.82	4.71	-3.12
3–5 mil	3.87	4.23	-1.25
5–10 mil	5.98	3.36	2.81
10–20 mil	7.62	2.42	3.58
> 20 mil	9.20	1.65	7.38
Panel B. After the parallel OTC transition (Jul 2006–Dec 2006)			
< 0.5 mil	3.57	6.23	-5.41
0.5–1 mil	4.25	5.64	-1.38
1–3 mil	5.19	4.18	1.60
3–5 mil	6.16	3.67	3.09
5–10 mil	7.11	2.56	5.98
10–20 mil	8.35	1.54	6.42
> 20 mil	9.57	0.68	8.75

5.4 Figures 5 and 6: single versus multiple price functions

Read:

- Figures 5 and 6 plot one-way relative effective spreads by venue, trade size, and bank size during and after the adjustment period.
- In both figures, the LOB is represented by one aggregate upward-sloping curve.
- In the OTC, there are three distinct downward-sloping curves:
 - **large banks** have the lowest spreads,
 - **medium banks** are in the middle,
 - **small banks** face the highest spreads.

- The separation across bank classes is economically large, especially for big trades. This is visual evidence for multiple OTC price functions and a single LOB price function.
- The post-transition figure shows that these patterns persist after the market settles into its new structure.

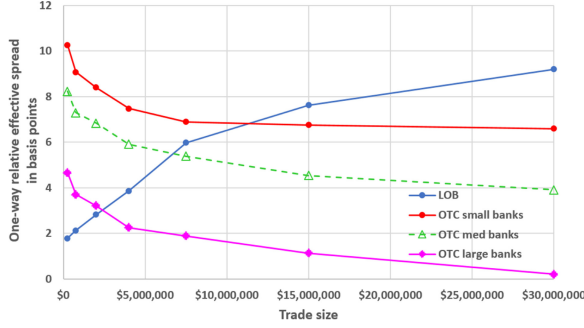


Fig. 5. Empirical one-way relative effective spread in basis points by venue, bank size, and trade size: during the adjustment (Jan 2006–Jun 2006).



Fig. 6. Empirical one-way relative effective spread in basis points by venue, bank size, and trade size: after the adjustment (Jul 2006–Dec 2006).

5.5 Table 5 and Table 6: selection correction and within-venue pricing

Read:

- Table 5 is the first-stage Probit for choosing OTC versus LOB. The coefficients say:
 - **medium banks** are significantly more likely than small banks to choose OTC (**0.59**),
 - trade-size coefficients rise strongly with size, from **0.39** for \$0.5–\$1 million to **2.93** above \$20 million,
 - the **transition dummy** is negative (**-0.86**), meaning OTC usage is lower while traders are still learning the new institution,
 - **buys** are more likely to choose OTC than sells.
- Table 6 is the main multivariate evidence on price functions.
 - In the **OTC** regression, the coefficients on **Medium** and **Large** are **-2.47** and **-3.98** bps, both highly significant. This means larger banks receive better OTC prices.
 - In the **LOB** regression, the corresponding coefficients are statistically insignificant. This means there is a single LOB price function.
- The trade-size coefficients in Table 6 imply:
 - a **downward-sloping** OTC price function over the relevant size range,
 - an **upward-sloping** LOB price function over the relevant size range.
- The inverse Mills ratio is significant in the OTC regression, confirming that venue selection matters empirically.
- These two tables are the strongest evidence that the core patterns are *not* artifacts of raw sorting.

5.6 Figures 7 and 8: fitted transaction-cost curves after correcting for selection

Read:

- Figure 7 plots the predicted OTC effective spread against trade size for small, medium, and large banks. All three curves slope downward, and larger banks always face lower costs.
- Figure 8 plots the predicted LOB effective spread against trade size. The LOB curve is upward-sloping, and liquidity worsens after the transition as trading migrates out.
- These fitted figures show that the raw visual facts survive in the regression-adjusted environment.

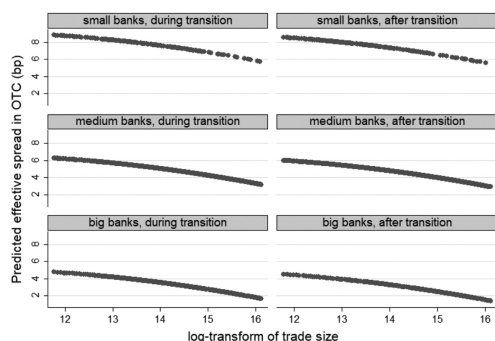


Fig. 7. One-way relative effective spread predicted by the Heckman two-stage for the OTC market; by bank size and trade size, during and after the transition. Note: The bottom 1 percentile and top 1 percentiles of trade size values are eliminated from the range of trade sizes. Log-transform of trade size ranging from 11.92 to 16.12 is equivalent to the trade size moving from \$0.15 million to \$10 million.

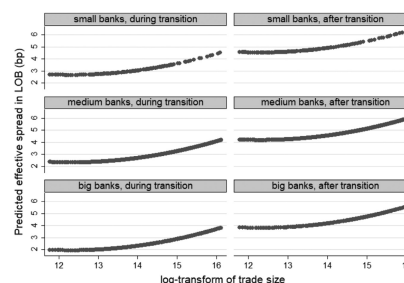


Fig. 8. One-way relative effective spread predicted by the Heckman two-stage for the LOB market; by bank size and trade size, during and after the transition. Note: The bottom 1 percentile and top 1 percentile of trade size values are eliminated from the range of trade sizes. Log-transform of trade size ranging from 11.92 to 16.12 is equivalent to the trade size moving from \$0.15 million to \$10 million.

5.7 Figure 9 and Table 7: theory meets the data

Read:

- Figure 9 is the model's main picture: one weakly upward-sloping step function for the LOB and three downward-sloping OTC curves for small, medium, and large banks.
- The crossing points of those curves yield the key prediction: traders with more bargaining power switch to OTC at smaller sizes.
- Table 7 tests that prediction using the spread difference **LOB minus OTC**. During the transition:
 - **large banks** switch around **\$3 million**,
 - **medium banks** switch around **\$10 million**,
 - **small banks** switch around **\$20 million**.
- After the transition, the switching thresholds fall further:
 - **large banks**: about **\$0.5 million**,
 - **medium banks**: about **\$5 million**,
 - **small banks**: about **\$10 million**.
- This is a sharp and intuitive confirmation of the model's comparative statics.

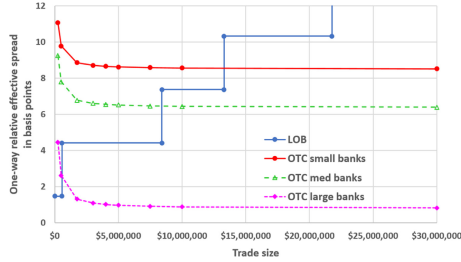


Fig. 9. A theoretical model of parallel LOB and OTC markets.

Table 7

The relative effective spread difference (LOB - OTC) by trade size and by bank size clienteles.

This table shows the relative effective spread difference (Limit Order Book - Over The Counter Market) in basis points (bps) for Chinese Yuan trades against the US dollar in the Chinese Interbank Foreign Exchange market. The spread difference is broken out by trade size and by bank size clienteles. All computations are based on China Foreign Exchange Trading System (CFETS) intraday transaction data from January 2006 to December 2006. *, **, and *** mean the 10%, 5%, 1% level of significance, respectively, based on the *t*-test of the difference between LOB and OTC.

	Venue difference (LOB minus OTC)		
	Small banks (bps)	Medium banks (bps)	Large banks (bps)
Panel A. During the parallel OTC transition (Jan 2006–Jun 2006)			
< 0.5 mil	–7.32***	–5.87***	–2.70***
0.5–1 mil	–6.22***	–4.87***	–1.66***
1–3 mil	–5.14***	–3.99***	–0.76***
3–5 mil	–3.56***	–2.51***	0.81**
5–10 mil	–2.23***	–1.34**	1.76***
10–20 mil	–1.28**	0.18**	3.29***
> 20 mil	1.60**	3.03**	5.17***
Panel B. After the parallel OTC transition (Jul 2006–Dec 2006)			
< 0.5 mil	–5.02***	–3.79***	–0.42***
0.5–1 mil	–3.96***	–2.73***	0.40**
1–3 mil	–3.28***	–1.77***	1.46***
3–5 mil	–1.57***	–0.41**	2.90***
5–10 mil	–0.22**	0.88***	4.10***
10–20 mil	0.85***	2.12***	5.54***
> 20 mil	3.74***	5.10***	7.24***

6 Short summary

- **Method.** The paper studies a rare market-structure reform in which a decentralized OTC market is introduced alongside a centralized LOB, and combines reduced-form evidence, selection correction, and a structural model of parallel markets.
- **Empirical facts.** Most trading migrates to OTC, especially large trades. The LOB price function is upward-sloping, the OTC price function is downward-sloping, and only the OTC exhibits multiple price functions across trader clienteles.
- **Theory.** Anonymous LOB trading creates adverse selection and therefore an upward-sloping price function; non-anonymous OTC trading with fixed processing costs and bargaining creates a downward-sloping price function and clientele-specific pricing.
- **Main implication.** Traders with stronger bargaining power switch to OTC at smaller sizes, and the data support that prediction.
- **Big lesson.** The same asset can generate very different transaction-cost schedules under different trading mechanisms. Market design and bargaining power, not just asset class, shape liquidity.

7 Conclusion

7.1 Core conclusion

- The introduction of a parallel OTC market fundamentally changed the Chinese interbank FX market.

- The OTC, not the LOB, became the dominant venue.
- The slope of the price function depends primarily on the **trading mechanism**, not on the fact that the asset is foreign exchange.
- The anonymity of the LOB leads to one common price function, whereas non-anonymous OTC bargaining leads to multiple price functions.

7.2 Economic intuition

- In an anonymous LOB, liquidity suppliers worry that larger incoming orders are more likely to be informed. They therefore charge more as size rises.
- In an OTC market, fixed processing costs are spread over more units as trade size increases, so per-unit cost falls with size.
- Because OTC counterparties know who they are trading with, bargaining power matters. Large banks can extract better prices than small banks.
- Putting the two venues side by side naturally implies size-dependent venue choice: small trades may stay on the LOB, while larger and more powerful traders move OTC.

7.3 Takeaway

This paper shows very clearly how *market structure* shapes liquidity. The Chinese interbank FX reform creates an unusually clean comparison between an anonymous centralized market and a non-anonymous decentralized market. The result is striking: the OTC dominates, large traders benefit most, and the same asset exhibits completely different price schedules across venues. The paper's theory then ties these facts together in a simple and convincing way.